Welcome!

Have you ever felt like your training wasn't producing results?

If yes! Meet Apex

Your AI-powered personal trainer

Apex, maximizing your training

- **Aaron Ricca**
  - Electrical and computer engineering
  - Ex professional skier
  - Climber
- **Derya Özsoy**
  - Informatik
  - Kickboxing

What's the problem ?

- training is far from optimized
- hard to find good exercises
- time is always too little

How does Apex work?

- the End User enters his **data**
- Apex learns with the collected **User data** and **data** from Internet and tries to find common patterns.
- At the end the User receives a personalized training plan

Data time !

Input attributes:

- Objective
- Available time
- Favorite exercises
- Biological attributes (Height, body fat%, medical condition, etc.)
- Alimentation (Cuisine, Diet type, enhancing substances, etc.)

Output attributes:

- Training periodization
- List of exercises
- Alimentation improvement tips
- Corresponding improvement % [Continuous]

Data Sources 📊

Research & Literature 📚

- Scientific papers
- Studies & publications
- Medical research

Real-World Data 💪

- Gyms & fitness centers
- Professional coaches
- Testers & athletes

User-Generated 📱

- Surveys & questionnaires
- Smartwatch data
- Fitness app tracking

Continuous Growth 🔄

- Data collected from users

Data Preprocessing 🔧

- Convert different units & scales to SI units
    - Height (feet → cm), Weight (lb → kg), etc.
- **Standardization** → for classification algorithms (KNN, Random Forest)
- **Normalization** → for regression algorithms

Few data?

→ More data?

🧪 Synthetic data!

SMOTE

***S**ynthetic **M**inority **O**versampling **Te**chnique*

- Used for **imbalanced datasets**
  → e.g., 90% fit people and 10% overweight
- Problem: ML models **bias the majority class**
- SMOTE **creates new synthetic data** of the minority class

How SMOTE Works

Our scenario: 10% overweight, 90% fit

-**User A:** Age 35, Weight 95kg, Beginner level

-**User B:** Age 40, Weight 100kg, Beginner level

SMOTE finds two **similar overweight users** and creates a new realistic profile:

-**Synthetic user C:** Age 37, Weight 97kg, Beginner level ✨

✅ Makes the minority class **denser and more balanced** (27%)

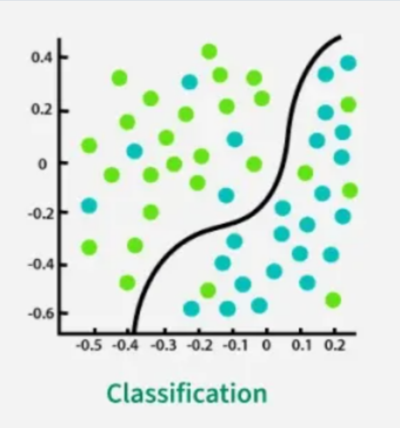
What are we using for Apex

- **Supervised Learning**
  - trained using labeled training data
  - learns to recognize patterns in order to make predictions for new, unknown and similar data
  - All Data are structured
    - Classification
    - Regression

Tools

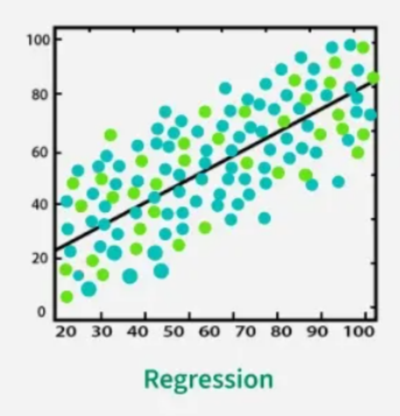
Classification 🎯

focuses on sorting observations into specific class
→ exp: We categorize some Data "*sick*" an "*healthy*" so that the user receives a training plan that does not harm his health



Regression 🎯

Based on making prediction or identification of trends
→ exp: predict a good training plan for the User



Algorithms 📈

- **Linear Regression**
  - Predict improvement percentages
  - Estimate workout outcomes
- **K-Nearest Neighbors (KNN)**
  - Find similar user profiles
  - Recommend based on comparable cases
- **Random Forests**
  - Robust decision-making
  - Handles complex non-linear relationships
  - Gradient Boosting

Data Processing 🔧

- **SMOTE** - Balance underrepresented groups
- **Cross-Validation** - Ensure model generalization
- **PCA (Principal component analysis)** - Reduce the data dimensionality

Why is Apex better?

- **Open source** - Transparent algorithms, community improvements
- **Data-driven optimization** - Personalized vs generic programs
- **Lower injury risk** - Biomechanical constraints and progressive overload
- **First of its kind** - Novel ML-based personal training approach
- **Continuous learning** - Improves with more user data

Possible risks, barriers and obstacles

- **Incomplete data can lead to incorrect results.**
  - can lead to injuries due to incorrect training suggestions
- **Time and resources.**
  - Data must always be updated and maintained.
  - Possible data loss.
  - High costs und delays.
- **Data breach.**

Ethical Considerations

- **Privacy & GDPR compliance** - Sensitive biometric and health data
- **Labor market impact** - Potential displacement of personal trainers
- **Economic effects** - Reduced gym memberships
- **Algorithmic bias** - Training data skewed toward young, fit persons
  - Risk: Inappropriate recommendations for underrepresented groups
  - Can lead to injuries in elderly or different body types
- **Dual-use concerns** - Technology repurposed beyond original intent
  - Example: Military training

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Apex: Any questions?

We're happy to dive deeper.